

# Assessing the Spatial Distribution of Visual Attention in a Virtual Environment: Development and Validation of a Novel VR-based Attentional Visual Field (AVF) Task

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## ABSTRACT

Visual attention is critical for everyday task performance and safety. The Attentional Visual Field Task (AVF) is an established, computerized method for assessing the distribution of visual attention across a wide visual field. High-fidelity virtual reality (VR) presents an opportunity for more ecological methods for assessing and training visual attention; however, this novel approach has not been examined. We developed a new VR-based AVF task, AVF-VE, using a Head-Mounted Display (HMD) VR device with an integrated eye-tracker, and conducted a study to validate this newly developed visual attention task. We further examined how visual attention is distributed in a virtual visual field. The findings suggest that the VR-based visual attention task is a valid and useful tool that can be used for future attention research and training. Unique characteristics of the spatial distribution of visual attention in the virtual environment observed in the current evaluation study are discussed.

## CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms.

## KEYWORDS

Virtual reality, Head-mounted display, Visual attention, Attentional visual field

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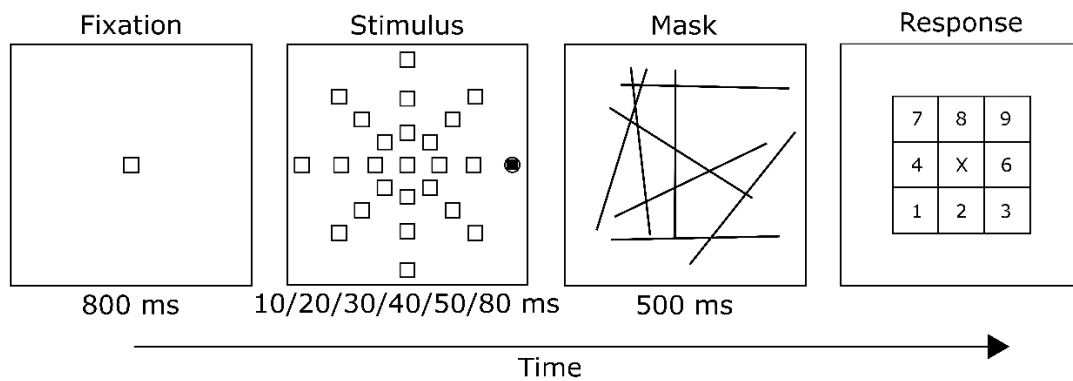
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## 1 INTRODUCTION

### 1.1 Visuospatial attention and the Attentional Visual Field (AVF) task

Visual attention is the ability to select relevant visual stimuli to process while filtering out irrelevant stimuli that will not be processed [18]. This mechanism is critical for daily functioning as the number of stimuli and amount of related information in any environment exceeds any person's immediate, limited processing capacity [5, 10, 21]. Visual attention is central to both task performance and safety. Visual attention allows people to detect and process critical visual objects, information, and potential hazards in an environment to respond to. The attentional visual field refers to the size of the visual field across which a person can process information successfully without performing eye movements [13]. Attentional processing of stimuli in the peripheral regions of vision has been consistently found to be slower and more error-prone than that of stimuli in the central visual region [6, 7]. And, reductions in the size of the attentional visual field are linked to a variety of everyday task performance and accident risks, especially among older adults, due to their inability to detect and process information in the visual periphery [20, 24]. For instance, declines in an attentional visual field are associated with increased crash risk among older drivers [1, 8, 9]. Also, the size of the attentional visual field is associated with fall risks that come from bumping into obstacles [4].

An individual's ability to distribute visual attention in space can be assessed by a computerized task that measures an ability to detect and respond to visual targets in a wide area of one's focal and peripheral visual field. For instance, the Attentional Visual Field (AVF) task [11, 12], or the Useful Field of View (UFOV) task [2, 19] that uses a similar paradigm, is a computerized task that displays a visual stimulus on a computer screen, and people need to identify a target among distractors while attending to a large visual field (Figure 1). For the AVF task, participants are placed in a fixed position, such as placing their head in a chinrest and sitting in front of a screen, and they are instructed to look at a gaze point. This is to ensure that the size and location of the stimuli are consistent; thus, visuospatial attention can be measured rigorously. The AVF task has been validated to map the spatial distribution of visual attention and used as a research, assessment, and training tool for visual attention abilities [12, 18, 23, 26]. For example, AVF performance was associated with crash risks as visuospatial attentional ability is critical to quickly notice and respond to hazards [1, 8, 9, 19].



**Figure 1:** An illustration of the computer-based AVF task procedure. This graphic was re-created based on the figure from [11].

## 1.2 Virtual Reality (VR) Technology

**1.2.1 Visual attention in a virtual environment.** Virtual Reality (VR) provides its users with an immersive virtual environment (VE) that can be designed to simulate a multitude of physical environments [17]. The rapid growth in VR adoption that has taken place in the past decade may have resulted from the significant improvement of a particular type of VR device, the Head-Mounted Display (HMD). HMD VR devices are compact, provide high-fidelity graphics, and use a display that shows computer-generated images to each eye and allows for physical environments and objects to be experienced virtually with limited physical risk to users. Most commercially available HMD VR headsets have a field of view (FOV) with restricted sizes and unnatural shapes compared to the natural human FOV. For example, VR headsets that are widely used in the current market have FOVs between 80 and 120 degrees diagonally, with vertical and horizontal FOVs being even lower, which is a significant reduction compared to human FOV of approximately 180 degrees [14–16]. The restricted FOV size and shape of VR headsets may affect visual attentional processing in a virtual environment. Some research has examined human visual behaviors and performance within VR, including saliency [25] and attentional orienting [3], however, the differences in visual attention that arise from using virtual reality have not been fully analyzed.

**1.2.2 VR-based visual attention assessment and training.** Technological advances of VR provide new opportunities to examine attentional performance in various environments and conditions. Compared to a computerized cognitive task, VR simulations have the potential for assessing cognitive performance in a more natural setting without restricting the test participant's movement or behaviors while maintaining the same rigorous procedures. For instance, if HMD VR is used to assess visual attention, participants do not need to be in a fixed position, such as being placed in a chinrest in front of a computer, because of the capability to display visual stimuli in a virtual visual field via the headset. Without their head stuck in a chinrest, participants can move their head and body around in the virtual environment, allowing for attention to be assessed in ways that more closely resemble how attention is employed in everyday life or specific scenarios of interest. The virtual environment that participants are placed into can be designed to

match physical environments that may present challenges for data collection. Dual-task paradigms can even be implemented effectively as participants no longer need to be in a fixed position. In addition, VR devices with integrated eye-trackers have the capability to present stimuli in a way that follows the participant's gaze, ensuring that the stimuli are presented at consistent visual angles and eccentricities based on their immediate gaze. Furthermore, VR can also be used as a training and intervention tool to improve visual attention functions and associated behaviors. Computerized AVF and UFOV tasks have been used as an effective intervention to reduce visuospatial attention deficits [2, 23]. VR-based visual attention tasks have the potential to effectively train visual attention, which may improve attention-related safety behaviors such as driving. People can use VR to receive training and assess its impact on their visual attention while they are in virtual environments that resemble physical environments, such as driving a car.

Despite the potential benefits, there could be unknown effects of using VR for assessing and training visual attention. Although the ability for VR-based research and training to place participants in specific virtual environments is presented as a major benefit of the VR modality, it is likely that this novel approach impacts task performance. Specifically, computer-based visual attention tasks are often overlaid on static backgrounds, such as a white or gray colored image, but in a VR version of these tasks, participants may be moving around in a virtual environment that has a variety of objects and backgrounds that could introduce some difficulty to the attention task. No research has been conducted that assesses the difference in performance between these two settings. Additionally, VR may impose its unique constraints on visual attentional processing due to restricted FOV and accommodation-convergence conflict. Prior research suggests that reduced FOV leads to inaccurate distance perception in VR [16, 27, 28], and it is likely that it has significant effects on visual attention; however, no research has examined what precise effects it may have. Finally, the accommodation-convergence conflict that occurs during the use of HMD VR headsets, where stimuli closer than two meters encounter visual perceptual problems [16], has not been explored in the domain of visual attention tasks.

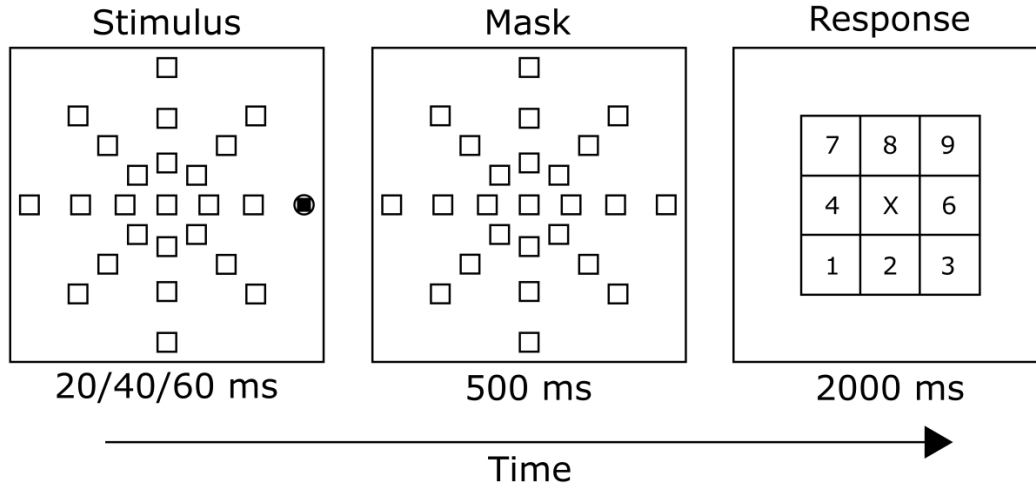


Figure 2: Illustration of a single trial of the VR-AVF task.

## 2 DEVELOPMENT OF THE AVF-VE TASK

We developed a novel VR-based AVF task, AVF-VE, based on the task procedure of the original computerized AVF task [11] (Figure 1), which can be used as an assessment and training tool for visuospatial attention. A computer-based AVF task presents a target stimulus (e.g., a filled square inside a lined circle) at a specific location with varying eccentricities (e.g., 10, 20, and 30 degrees) and directions (e.g., eight locations), with or without distractors (e.g., lined squares). After the stimulus image disappears, a masking image composed of randomly oriented lines is displayed to remove the afterimage of the target stimulus, and then a response screen appears for participants to indicate the direction of the target stimuli by moving the cursor to and clicking on one of the response squares as quickly and accurately as possible [12]. The AVF-VE task was developed in Unity with the SteamVR plugin from Valve and matched to the computer-based AVF task paradigm. Within the VR task, the stimuli are displayed based on real-time gaze point data collected using an integrated eye tracker. We used the *hmd-eyes* package from Pupil Labs to detect real-time gaze point data. The task program was designed that if a participant's gaze point during the task is located in far peripheral visual areas, which would result in all or parts of the stimuli being cut off, the display of subsequent trials are delayed until the gaze point moves back to a less extreme area; thus, all 24 locations are displayed in the participant's visual field.

Similar to the computer-based AVF, for a single trial of AVF-VE, one target stimulus (e.g., a filled square inside a lined circle) appears at a particular location with varying eccentricities (e.g., 10, 20, and 30 degrees) and directions (e.g., eight locations), with or without distractors (e.g., lined squares) (Figure 2). After a masking image is presented, participants are asked to indicate the direction of the target stimuli on a keypad or a remote controller as quickly and accurately as possible. Both detection rate (i.e., response accuracy) and reaction time to the target stimuli are recorded. The virtual environment (Figure 3) was purchased from TurboSquid

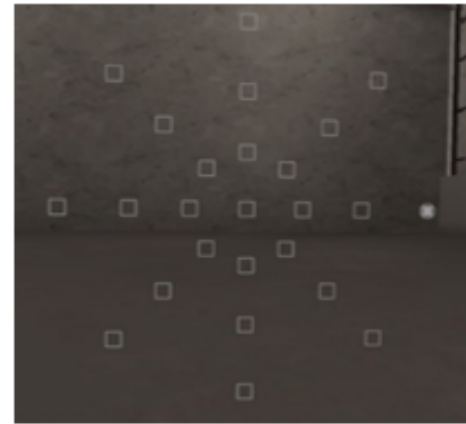


Figure 3: An example image of a target presentation of the AVF-VE task.

(<https://www.turbosquid.com/>) and modified by the researchers to maximize performance and to correct lighting and texture problems. The AVF-VE task is modifiable and can be presented within different virtual environments and can use varying virtual stimuli. Similar to the original AVF task, eccentricities of the stimuli, presentation times, number of repetitions, and response mechanisms can also be modified.

## 3 EVALUATION STUDY

We conducted a study to evaluate the newly developed AVF-VE empirically. In this study, we evaluated the validity of AVF-VE and expected to find similar effects that have been observed in previous computer-based AVF studies. We expected response accuracy to be lower for targets presented in peripheral visual areas than focal

visual areas and for targets presented for shorter durations. Our second aim was to evaluate visual attentional performance in a virtual space using AVF-VE. We examined if there are potential changes in the spatial distribution of visual attention when measured in a restricted virtual environment compared to the distribution when measured in a physical environment using a traditional computerized task.

### 3.1 Methods

**3.1.1 Participants.** We recruited a total of 56 undergraduate students; however, nine of those students were dismissed from the study early due to issues with calibrating the eye tracker. One additional student was dismissed due to symptoms of simulator sickness. Six participants' data were excluded from the data analysis because their accuracy was lower than 25%. We analyzed data from 40 participants (age: mean = 18.72,  $SD = 1.78$ ). Among the participants, ten were males, 29 were females. One person did not identify as male or female. Fifteen participants reported that they had prior experience with HMD VR headsets and two participants had experience with the Cave Automatic Virtual Environment (CAVE) VR. All participants reported having a normal or corrected-to-normal vision.

**3.1.2 AVF-VE Task and Equipment.** The AVF-VE task used in this study consisted of 288 trials with 72 trials in each of four blocks. The target stimulus was a dark-gray circle with a size of 2.7 degrees both vertically and horizontally surrounding a filled-in square with a size of 1.4 degrees in both dimensions. The distractors were unfilled dark-gray squares with a size of 2.7 degrees. A target stimulus appeared in one of the eight directions (See figure 2) and at one of the three eccentricities (i.e., 10, 20, and 30 degrees). The stimuli's exposure time or duration varied with the three levels being 20, 40 and 60 ms. The masking image was present for 500 ms, and then participants had up to two seconds to respond. Participants indicated the direction of the target stimulus on a wireless numeric keypad sized 5.11 x 3.34 x 1.57 inches. The keypad had all keys removed aside from 1, 2, 3, 4, 6, 7, 8, and 9.

The HMD VR headset used to conduct the current study was an HTC Vive Pro with a 110-degree maximum diagonal FOV and a 90 Hz refresh rate. The Pupil Labs HTC Vive add-on eye-tracker was used and had a 200 Hz sampling rate. Both response accuracy and reaction time to the target stimuli were recorded. The data that we analyzed in the current study focused on the response accuracy for the AVF task. We also collected the total task time and walking task response rate; however, these data were not included in the current study.

**3.1.3 Experiment Procedure.** All participants gave their informed consent to participate and completed a demographic survey. Participants then wore the HMD VR headset and completed the AVF-VE task while performing a secondary walking task, which consisted of them simply walking to one of the three markers in the virtual environment that were randomly selected and turned into a green color. This secondary task was added to the current study to modify the AVF-VE task to be a more ecologically valid measure of visual attention by having participants engage in an everyday task of walking while their visual attentional performance was being

assessed. Participants held a wireless number pad and responded by pressing a corresponding key for the direction where the target stimulus appeared. If any participants experienced simulator sickness during the experiment, the experiment was immediately stopped, participants were given a chair to sit in to recover, and once they felt well again the participants were allowed to leave without penalty. We administered a computer-based AVF as part of the current experiment; however, the data collection encountered an unexpected technical issue and so this portion of the data was incomplete.

### 3.2 Results

We analyzed the AVF-VE task performance to evaluate the validity of the novel AVF-VE task and examine the attentional distribution in a virtual space.

**3.2.1 AVF-VE Validity.** To determine whether there were differences across the three exposure times of 20 ( $M = .36$ ,  $SD = .12$ ), 40 ( $M = .51$ ,  $SD = .14$ ), and 60 ms ( $M = .57$ ,  $SD = .13$ ), a repeated measures ANOVA was conducted (Figure 4). Increased accuracy was expected when a stimulus was presented for a longer period of time. A significant effect of exposure time on response accuracy was observed,  $F(2, 37) = 26.84$ ,  $p < .05$ ,  $\eta^2 = .31$ , and post-hoc t-tests were conducted to determine where the differences in accuracy existed. The analysis indicated that accuracy was lower with 20 ms of exposure time when compared to both 40,  $t(39) = 12.02$ ,  $p < .05$ ,  $d = 1.12$ , and 60 ms,  $t(39) = 14.36$ ,  $p < .05$ ,  $d = 1.68$ . Additionally, lower accuracy was observed with 40 ms when compared to 60 ms,  $t(39) = 5.68$ ,  $p < .05$ ,  $d = .45$ .

We also conducted a repeated measures ANOVA (Figure 5) to examine the differences in AVF-VE performance across the three eccentricities of 10 ( $M = .51$ ,  $SD = .15$ ), 20 ( $M = .51$ ,  $SD = .13$ ), and 30 degrees ( $M = .43$ ,  $SD = .13$ ). Decreased accuracy was expected with increasing eccentricity as it is more difficult to detect a visual stimulus appearing in the visual periphery compared to that in the visual fovea. A significant effect of eccentricity was observed,  $F(2, 37) = 4.65$ ,  $p < .05$ ,  $\eta^2 = .07$ . Post-hoc t-tests revealed that accuracy was higher at 10 degrees of eccentricity when compared to 30 degrees,  $t(39) = 3.85$ ,  $p < .05$ ,  $d = .56$ . Higher accuracy was also observed at 20 degrees of eccentricity when compared to 30 degrees,  $t(39) = 6.66$ ,  $p < .05$ ,  $d = .65$ .

**3.2.2 Attention Distribution in a Virtual Visual Field.** We examined how attention was distributed in a virtual visual field and also what differences in task accuracy may exist across regions of the virtual visual field. Figure 6 presents the AVF-VE accuracy across the eight visual field directions. Our results showed that the spatial distribution of visual attention in VR appeared to have a rectangular shape, with inferior vertical performance and superior diagonal performance.

We conducted 2 paired samples t-tests between the upper and lower regions of the visual field and between the left and right regions. The comparison between the upper and lower visual field regions showed that the accuracy of visual attentional processing was superior in the upper field ( $M = .50$ ,  $SD = .14$ ) compared to the lower field ( $M = .46$ ,  $SD = .14$ ),  $t(39) = 2.16$ ,  $p < .05$ ,  $d = .33$  (Figure 7). This result is consistent with previous studies examining

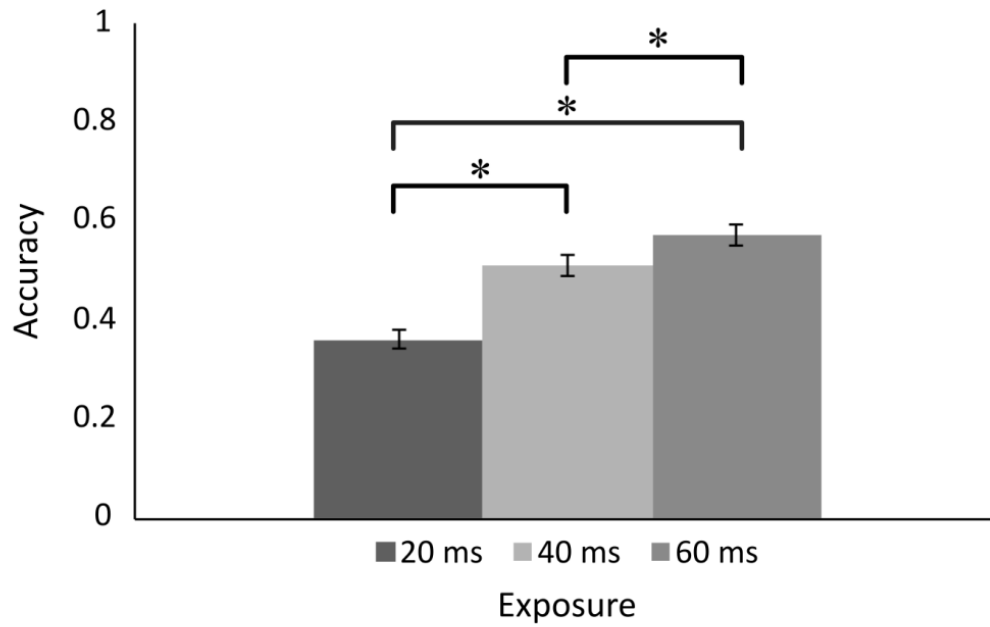


Figure 4: Comparisons of AVF task performance across exposure times. Note. \*  $p < .05$ , error bars represent  $\pm 1$  standard error.

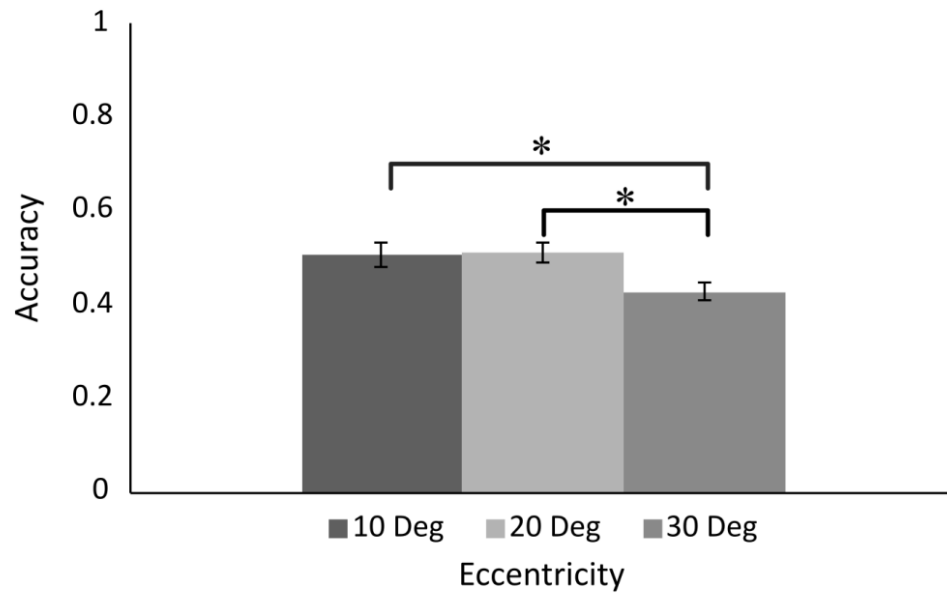
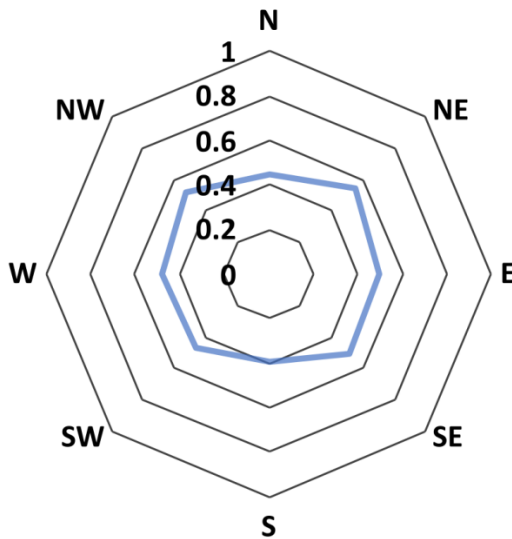
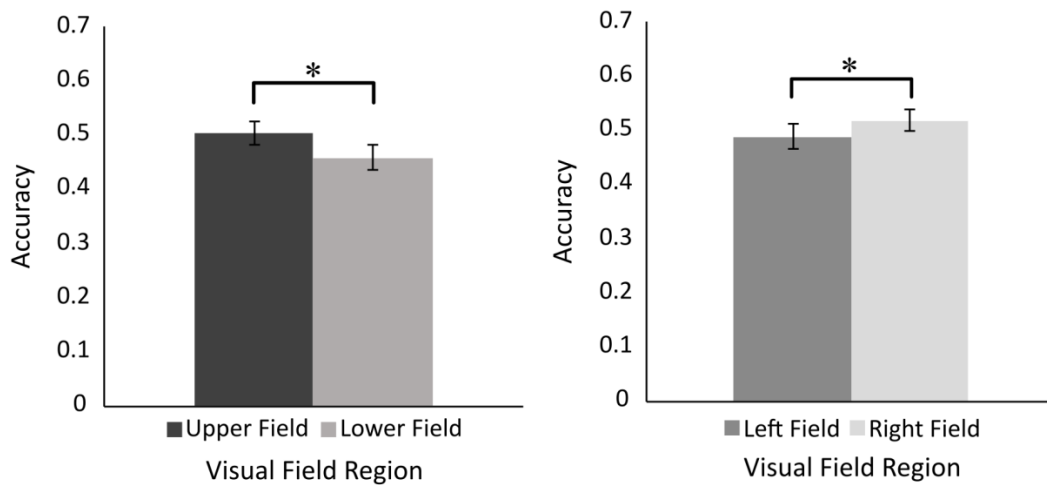


Figure 5: Comparisons of AVF task performance across eccentricities. Note. \*  $p < .05$ , error bars represent  $\pm 1$  standard error.



**Figure 6: Mean accuracy visualized across eight AVF-VE directions. Note. The marked point for each direction represents the direction's mean target detection rate.**

the spatial distribution of visual attention using a computerized AVF task [12, 22]. A novel effect was uncovered in the analysis examining the differences in AVF-VE accuracy between the left ( $M = .49$   $SD = .14$ ) and right visual field regions, ( $M = .52$   $SD = .13$ ) with the right visual field resulting in higher accuracy,  $t(39) = 2.16$ ,  $p < .05$ ,  $d = .22$ .



**Figure 7: Comparisons of AVF task performance between upper and lower visual field (left) and between left and right visual field (right). Note. \*  $p < .05$ , error bars represent  $\pm 1$  standard error.**

**3.2.3 AVF-VE and Secondary Task Performance Correlations.** A correlational analysis was conducted to determine if performance on the AVF-VE task was associated with the secondary walking task performance. The result indicated that the number of virtual markers reached in the secondary task was not correlated with AVF-VE accuracy,  $r(39) = .02$ ,  $p = .90$ , suggesting that the AVF-VE performance was not affected by a secondary physical task performance.

## 4 DISCUSSION AND FUTURE WORK

The findings from the current study evaluating the AVF-VE task suggest that this newly developed visual attention task is valid and useful to examine the spatial distribution of attention in virtual environments. The results observed were mainly consistent with past research conducted with computer-based AVF tasks, showing that as eccentricity increased response accuracy decreased, and as exposure time increased accuracy increased. Additionally, targets that appeared in the upper visual field were correctly detected more often than targets in the lower visual field, as has been observed in previous research. The consistent findings in this study show that the AVF task is transferrable to VR for measuring visuospatial attention.

Furthermore, we examined the distribution of visual attention in a virtual environment and observed a trend that participants more accurately responded to targets along the horizontal meridian and in the diagonal directions than targets along the vertical meridian. This led to a moderately rectangular distribution of visual attention across the visual field, as can be seen in Figure 7. A rectangular distribution of attention could indicate that the FOV constraints of the HMD VR device used in the current study impacted AVF task performance by shaping the AVF. In addition, there was one interesting result observed in the current study: targets in the right visual field had higher response accuracy than targets in the left visual field. To our knowledge, no previous studies investigating

visuospatial attention found significant differences between the left vs. right visual field. This result may have been observed due to the response mechanism of the current VR task. In our study, participants held a wireless number pad with both hands and were asked to respond using both hands depending on which direction the target appeared in. Almost all participants completed the task by responding with either their left or right thumb depending on the side of the visual field the target was observed in. Participants may have been more accurate at responding to targets in the right field due to the finer motor skills in their dominant hand, which is more likely a right hand than a left hand. However, we did not collect the dominant hands of the participants: thus, the current observation showing differences between left and right visual field regions may require additional investigations.

In a future study, we plan to directly compare the spatial distribution of visual attention between computer-based and VR-based AVF tasks. This planned study will allow us to understand at a greater depth how visuospatial attention is different in a virtual environment with the constraints of VR, such as the size of FOV. In addition to this planned study, we have also developed an AR-based AVF task. Future studies will compare AVF performance in VR and AR to understand how these two mediums affect visuospatial attention in an extended reality environment. These comparisons will also likely provide insight into how the FOV restriction of an HMD VR device impacts the shape of the AVF as AR devices used do not restrict a person's natural FOV.

## REFERENCES

- [1] K. Ball, C. Owsley, M. E. Sloane, D. L. Roenker, and J. R. Bruni. 1993. Visual attention problems as a predictor of vehicle crashes in older drivers. *Investig. Ophthalmol. Vis. Sci.* 34, 11 (1993), 3110–3123.
- [2] Karlene K. Ball, Bettina L. Beard, Daniel L. Roenker, Richard L. Miller, and David S. Griggs. 1988. Age and visual search: expanding the useful field of view. *J. Opt. Soc. Am. A* 5, 12 (1988), 2210. DOI:https://doi.org/10.1364/josaa.5.002210
- [3] Alexia Bourgeois, Emmanuel Badier, Naem Baron, Fabien Carruzzo, and Patrik Vuilleumier. 2018. Influence of reward learning on visual attention and eye movements in a naturalistic environment: A virtual reality study. *PLoS One* 13, 12 (2018), 1–16. DOI:https://doi.org/10.1371/journal.pone.0207990
- [4] Aimee Teo Broman, Sheila K. West, Beatriz Muñoz, Karen Bandeen-Roche, Gary S. Rubin, and Kathleen A. Turano. 2004. Divided visual attention as a predictor of bumping while walking: The Salisbury Eye Evaluation. *Investig. Ophthalmol. Vis. Sci.* 45, 9 (2004), 2955–2960. DOI:https://doi.org/10.1167/i0vs.04-0219
- [5] Marisa Carrasco. 2011. Visual attention: The past 25 years. *Vision Res.* 51, 13 (2011), 1484–1525. DOI:https://doi.org/10.1016/j.visres.2011.04.012
- [6] Marisa Carrasco and Irene Chang. 1995. The interaction of objective and subjective organizations in a localization search task. *Percept. Psychophys.* 57, 8 (1995), 1134–1150. DOI:https://doi.org/10.3758/BF03208370
- [7] Marisa Carrasco and Karen S. Frieder. 1997. Cortical magnification neutralizes the eccentricity effect in visual search. *Vision Res.* 37, 1 (1997), 63–82. DOI:https://doi.org/10.1016/S0042-6989(96)00102-2
- [8] HeeSun Choi, Daniel Grün, and Jing Feng. 2015. Self-reported attentional failures during driving relates to on-road crashes and simulated driving performance of older drivers. *Transp. Res. Board* 919 (2015), 16p. Retrieved from [http://www.jfenglab.com/uploads/1/1/9/7/119733837/choi\\_et\\_al\\_2015\\_trb15-5079.pdf%0A](http://www.jfenglab.com/uploads/1/1/9/7/119733837/choi_et_al_2015_trb15-5079.pdf%0A); <https://trid.trb.org/view/1339022>
- [9] Olivio J. Clay, Virginia G. Wadley, Jerri D. Edwards, David L. Roth, Daniel L. Roenker, and Karlene K. Ball. 2005. Cumulative meta-analysis of the relationship between useful field of view and driving performance in older adults: Current and future implications. *Optom. Vis. Sci.* 82, 8 (2005), 724–731. DOI:https://doi.org/10.1097/01.opx.0000175009.08626.65
- [10] Howard Egeth and Daniel Kahneman. 1975. Attention and Effort. DOI:https://doi.org/10.2307/1421603
- [11] Jing Feng, Fergus I.M. Craik, Brian Levine, Sylvain Moreno, Gary Naglie, and Hee Sun Choi. 2017. Differential age-related changes in localizing a target among distractors across an extended visual field. *Eur. J. Ageing* 14, 2 (2017), 167–177. DOI:https://doi.org/10.1007/s10433-016-0399-7
- [12] Jing Feng and Ian Spence. 2014. Upper visual field advantage in localizing a target among distractors. *Iperception*. 5, 2 (2014), 97–100. DOI:https://doi.org/10.1068/i0625rep
- [13] Shirin E. Hassan, Kathleen A. Turano, Beatriz Muñoz, Cynthia Munro, Karen Bandeen Roche, and Sheila K. West. 2008. Cognitive and vision loss affects the topography of the attentional visual field. *Investig. Ophthalmol. Vis. Sci.* 49, 10 (2008), 4672–4678. DOI:https://doi.org/10.1167/i0vs.07-1112
- [14] Jan Horský. 2020. Crowdsourcing VR headset data VR headset database. Retrieved from <https://www.infinite.cz/projects/HMD-tester-virtual-reality-headset-database-utility>
- [15] Jan Horský. 2020. VR field of view measured and explained. Retrieved from <https://www.infinite.cz/blog/VR-Field-of-View-measured-explained>
- [16] J. Adam Jones, J. Edward Swan, and Mark Bolas. 2013. Peripheral stimulation and its effect on perceived spatial scale in virtual environments. *IEEE Trans. Vis. Comput. Graph.* 19, 4 (2013), 701–710. DOI:https://doi.org/10.1109/TVCG.2013.37
- [17] Mel Slater Martin Usoh and Anthony Steed. 1995. Taking Steps: The Influence of a Walking Technique on Presence in Virtual Reality. *ACM Trans. Comput. Interact.* 2, 3 (1995), 201–219. DOI:https://doi.org/10.1145/210079.210084
- [18] S. A. McMains and S. Kastner. 2009. Visual attention. In *Encyclopedia of neuroscience*. 4296–4302.
- [19] Renee S. Myers, Karlene K. Ball, Thomas D. Kalina, David L. Roth, and Kathryn T. Goode. 2000. Relation of useful field of view and other screening tests to on-road driving performance. *Percept. Mot. Skills* 91, 1 (2000), 279–290. DOI:https://doi.org/10.2466/pms.2000.91.1.279
- [20] [1] Cynthia Owsley, Karlene Ball, and Dewanna M. Keeton. 1995. Relationship between visual sensitivity and target localization in older adults. *Vision Res.* 35, 4 (1995), 579–587. DOI:https://doi.org/10.1016/0042-6989(94)00166-J
- [21] Michael I Posner. 1980. Orienting or attention. *Q. J. Exp. Psychol.* 32, July 1979 (1980), 3–25.
- [22] Fred H. Previc. 1990. Functional specialization in the lower and upper visual fields in humans: Its ecological origins and neurophysiological implications. *Behav. Brain Sci.* 13, 3 (1990), 519–542. DOI:https://doi.org/10.1017/S0140525X00080018
- [23] Eric Richards, Patrick J. Bennett, and Allison B. Sekuler. 2006. Age related differences in learning with the useful field of view. *Vision Res.* 46, 25 (2006), 4217–4231. DOI:https://doi.org/10.1016/j.visres.2006.08.011
- [24] Allison B. Sekuler, Patrick J. Bennett, and Mortimer Mamelak. 2000. Effects of aging on the useful field of view. *Exp. Aging Res.* 26, 2 (2000), 103–120. DOI:https://doi.org/10.1080/036107300243558
- [25] Vincent Sitzmann, Ana Serrano, Amy Pavel, Maneesh Agrawala, Diego Gutierrez, Belen Masia, and Gordon Wetzstein. 2018. Saliency in VR: How Do People Explore Virtual Environments? *IEEE Trans. Vis. Comput. Graph.* 24, 4 (2018), 1633–1642. DOI:https://doi.org/10.1109/TVCG.2018.2793599
- [26] Jeremy M Wolfe, Patricia O'Neill, and Sara C Bennett. 1998. Why are there Eccentricity Effects in Visual Search? *Percept. Psychophys.* 60, 1 (1998), 140–156.
- [27] Bochao Li, Anthony Nordman, James Walker, and Scott A. Kuhl. 2016. The effects of artificially reduced field of view and peripheral frame stimulation on distance judgments in HMDs. *Proc. ACM Symp. Appl. Perception, SAP 2016* (2016), 53–56. DOI:https://doi.org/10.1145/2931002.2931013
- [28] Bochao Li, Ruimin Zhang, Anthony Nordman, and Scott A. Kuhl. 2015. The effects of minification and display field of view on distance judgments in real and HMD-based environments. *Proc. - SAP 2015 ACM SIGGRAPH Symp. Appl. Percept.* (2015), 55–58. DOI:https://doi.org/10.1145/2804408.2804427